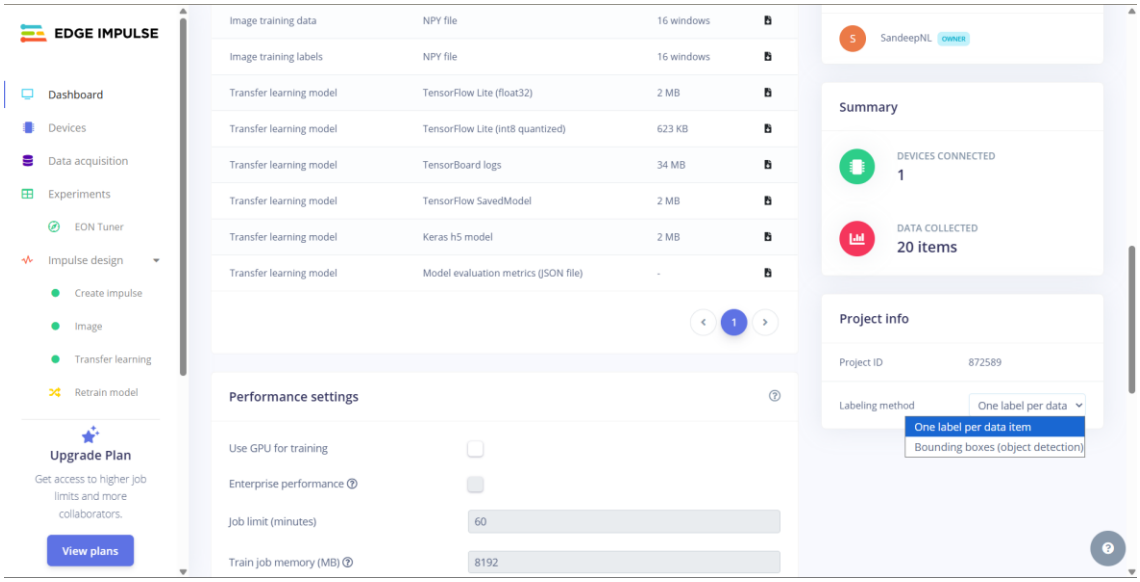
Edge Impulse Assignment

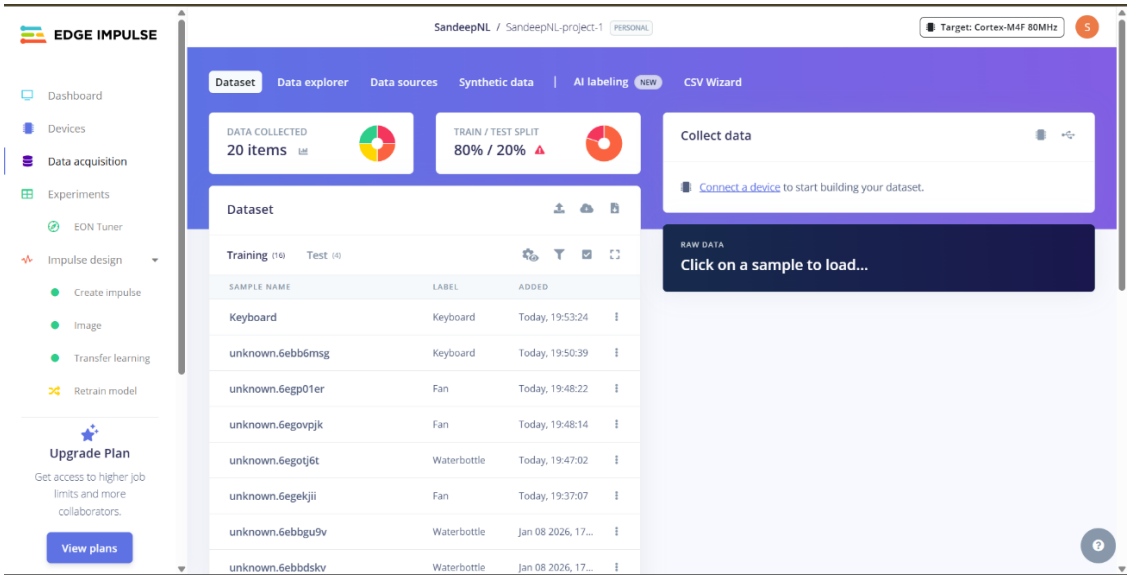
N. Lakshmi Sandeep

25MML0006

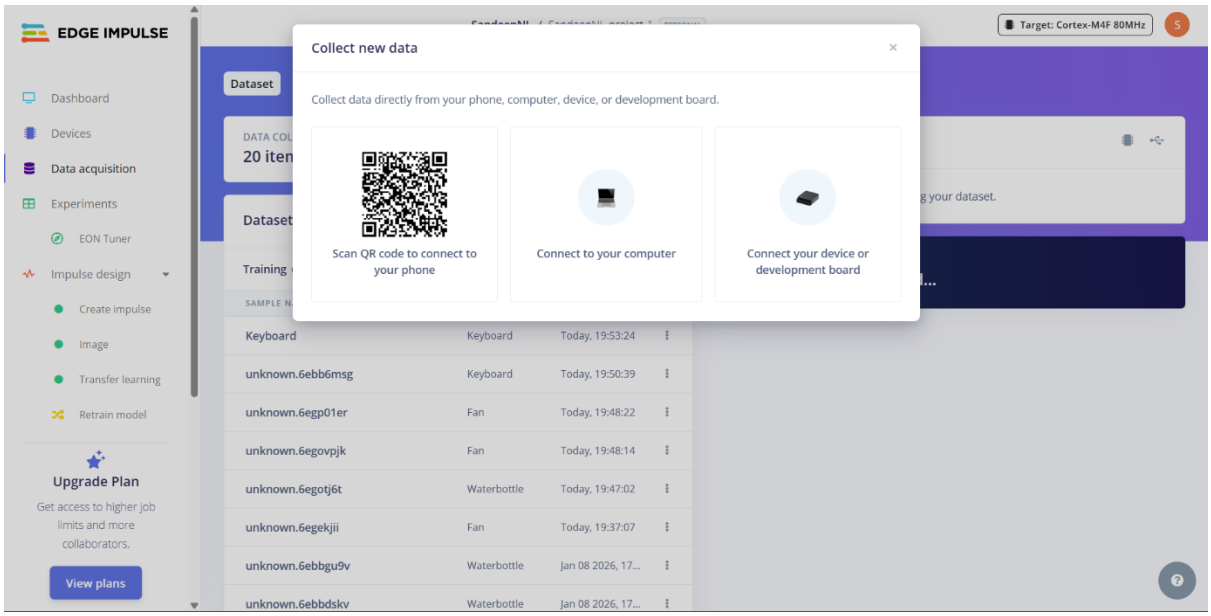
Capture images of different objects in different angles, label the names, upload the images into edge impulse website, train the images on any model and predict the accuracy of the output.



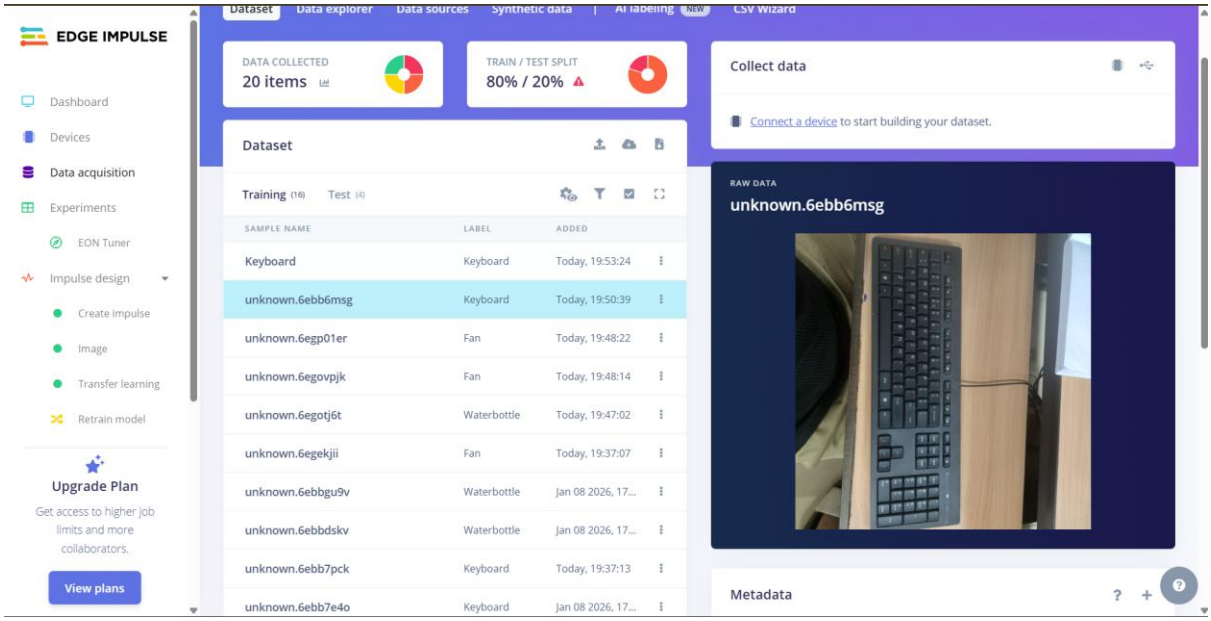
- In the project info we can choose any one of the labelling methods, if I take bounding method I have to detect the object specifically by selecting the image position.



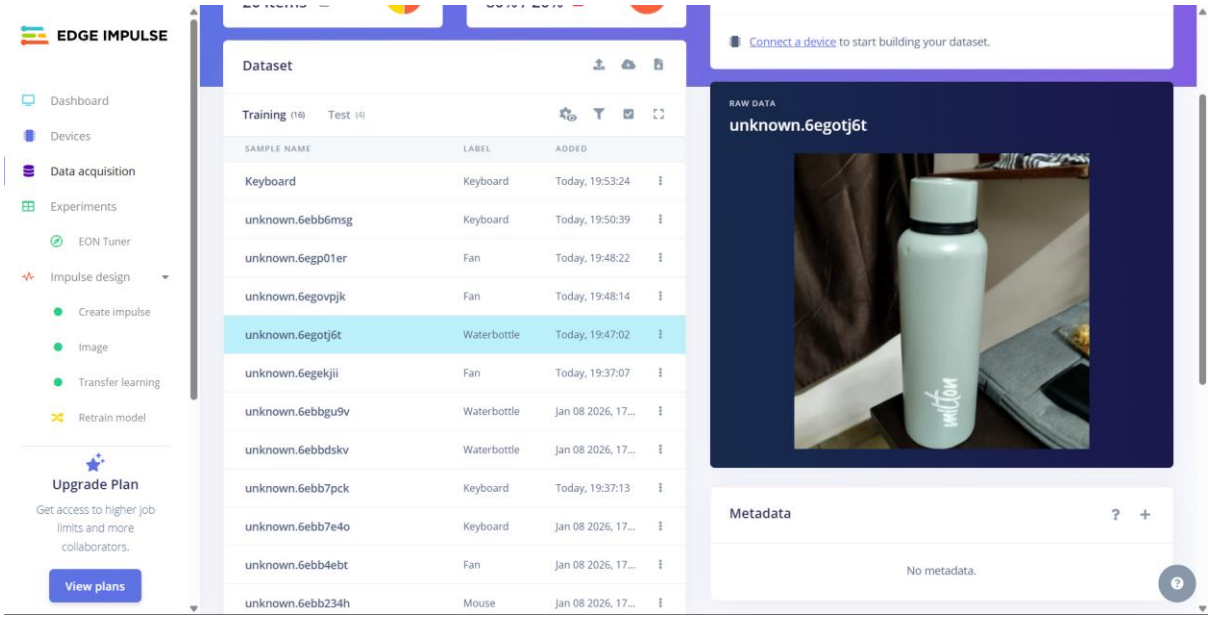
From the Data acquisition bar collect the data from either of those options


EDGE IMPULSE
Target: Cortex-M4F 80MHz
Dashboard
Devices
Data acquisition
Experiments
EON Tuner
Impulse design
Create impulse
Image
Transfer learning
Retrain model
Upgrade Plan
Get access to higher job limits and more collaborators.
View plans
Dataset
Collect new data
Collect data directly from your phone, computer, device, or development board.
Scan QR code to connect to your phone
Connect to your computer
Connect your device or development board
Keyboard Keyboard Today, 19:53:24
unknown.6ebb6msg Keyboard Today, 19:50:39
unknown.6egp01er Fan Today, 19:48:22
unknown.6egovpjk Fan Today, 19:48:14
unknown.6egotj6t Waterbottle Today, 19:47:02
unknown.6egekjii Fan Today, 19:37:07
unknown.6ebbgu9v Waterbottle Jan 08 2026, 17...
unknown.6ebbdskv Waterbottle Jan 08 2026, 17...

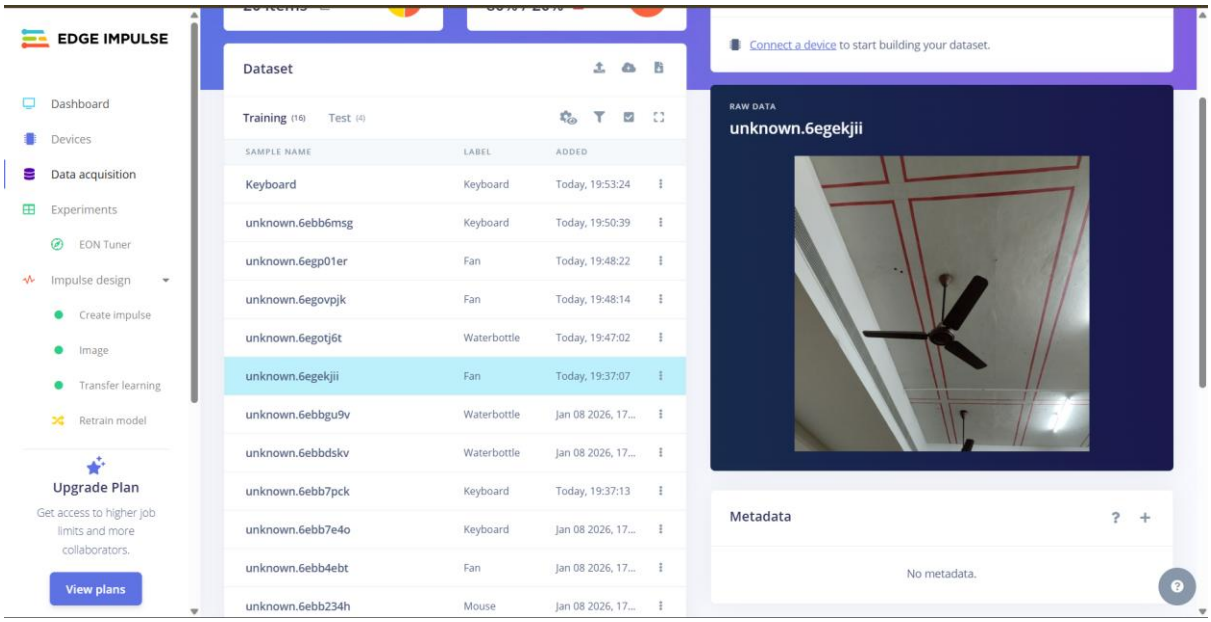
Keyboard:


EDGE IMPULSE
Dashboard
Devices
Data acquisition
Experiments
EON Tuner
Impulse design
Create impulse
Image
Transfer learning
Retrain model
Upgrade Plan
Get access to higher job limits and more collaborators.
View plans
Dataset
Data explorer
Data sources
Synthetic data
AI labeling NEW
CSV wizard
DATA COLLECTED
20 items
TRAIN / TEST SPLIT
80% / 20%
Collect data
Connect a device to start building your dataset.
Training (16) Test (4)
SAMPLE NAME LABEL ADDED
Keyboard Keyboard Today, 19:53:24
unknown.6ebb6msg Keyboard Today, 19:50:39
unknown.6egp01er Fan Today, 19:48:22
unknown.6egovpjk Fan Today, 19:48:14
unknown.6egotj6t Waterbottle Today, 19:47:02
unknown.6egekjii Fan Today, 19:37:07
unknown.6ebbgu9v Waterbottle Jan 08 2026, 17...
unknown.6ebbdskv Waterbottle Jan 08 2026, 17...
unknown.6ebb7pck Keyboard Today, 19:37:13
unknown.6ebb7e4o Keyboard Jan 08 2026, 17...
RAW DATA
unknown.6ebb6msg
Metadata

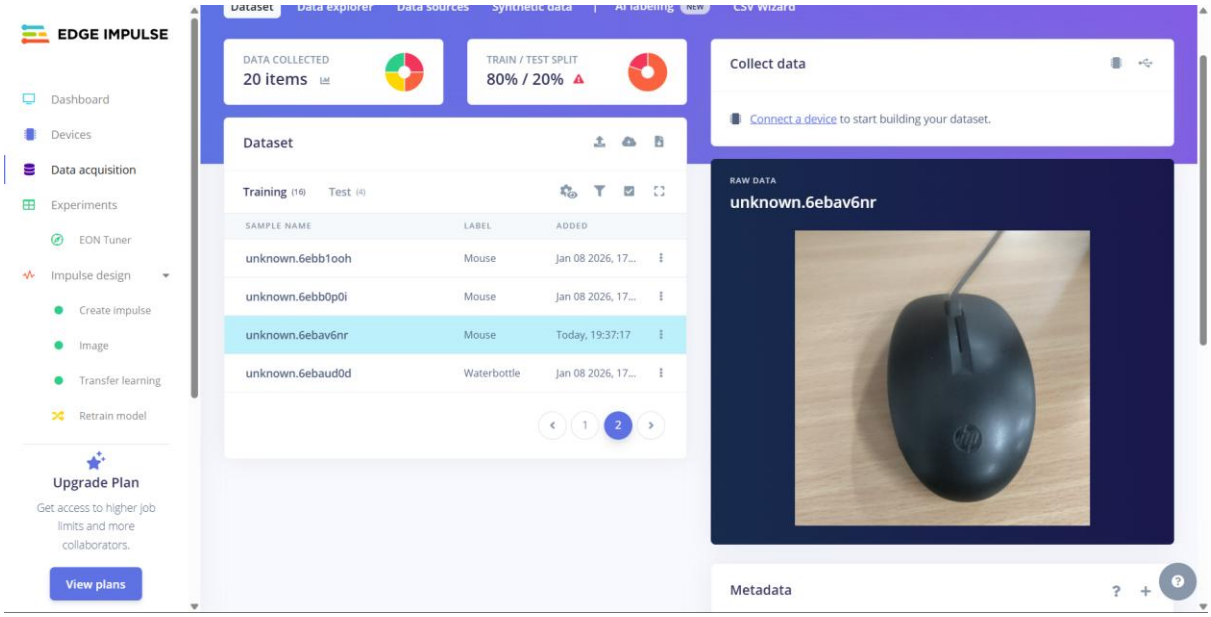
Water bottle



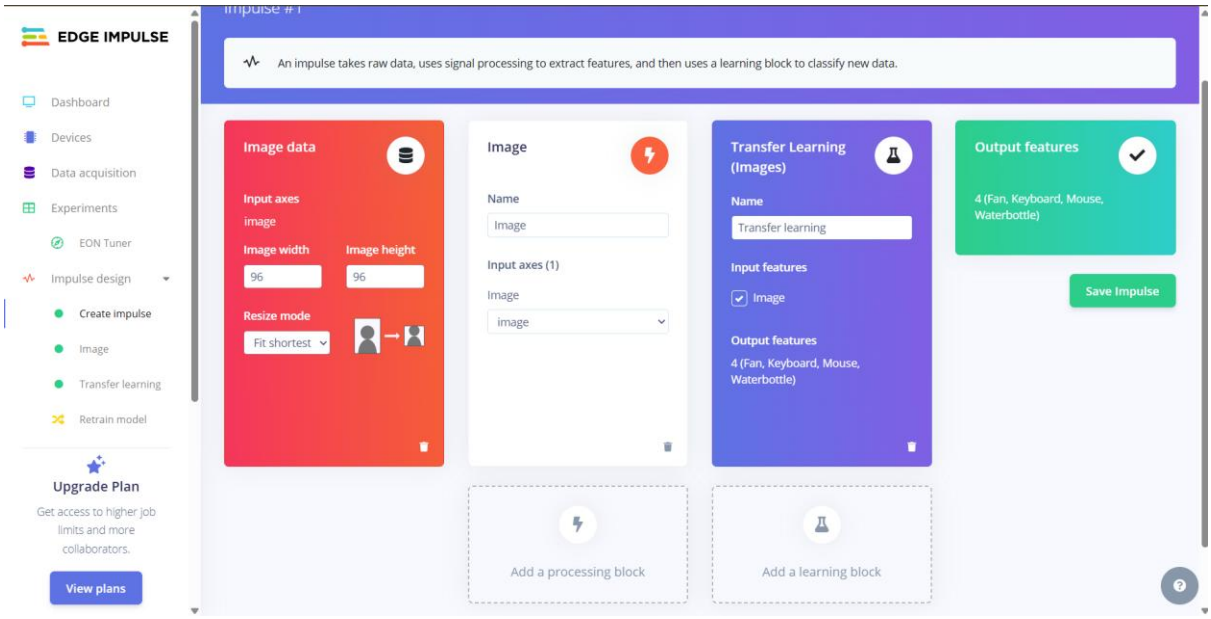
Fan



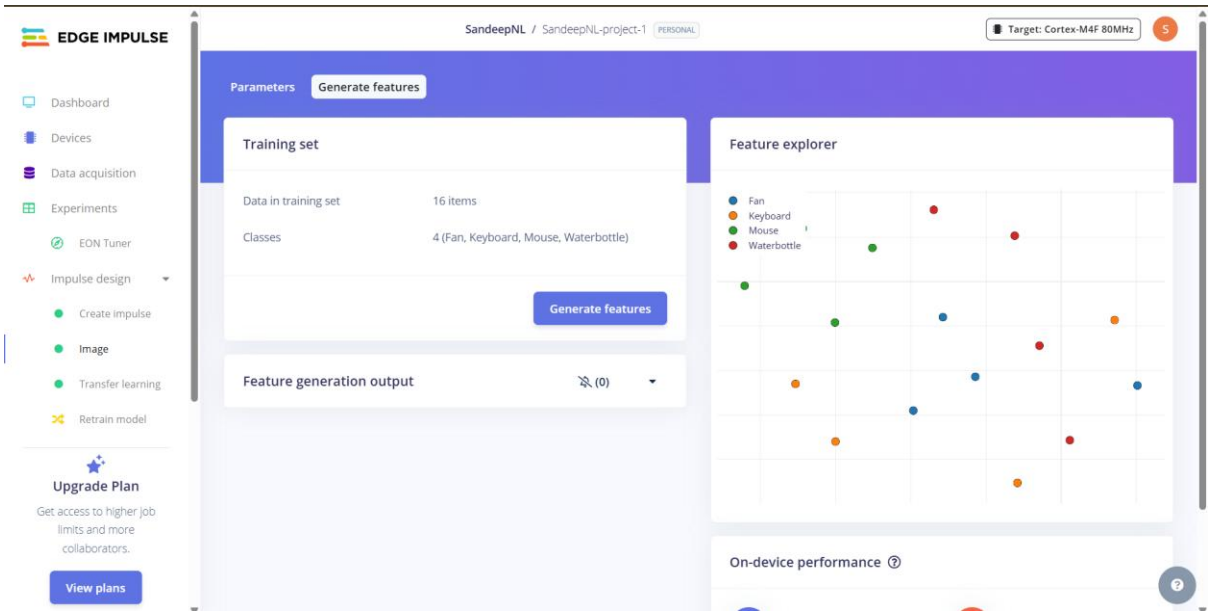
Mouse



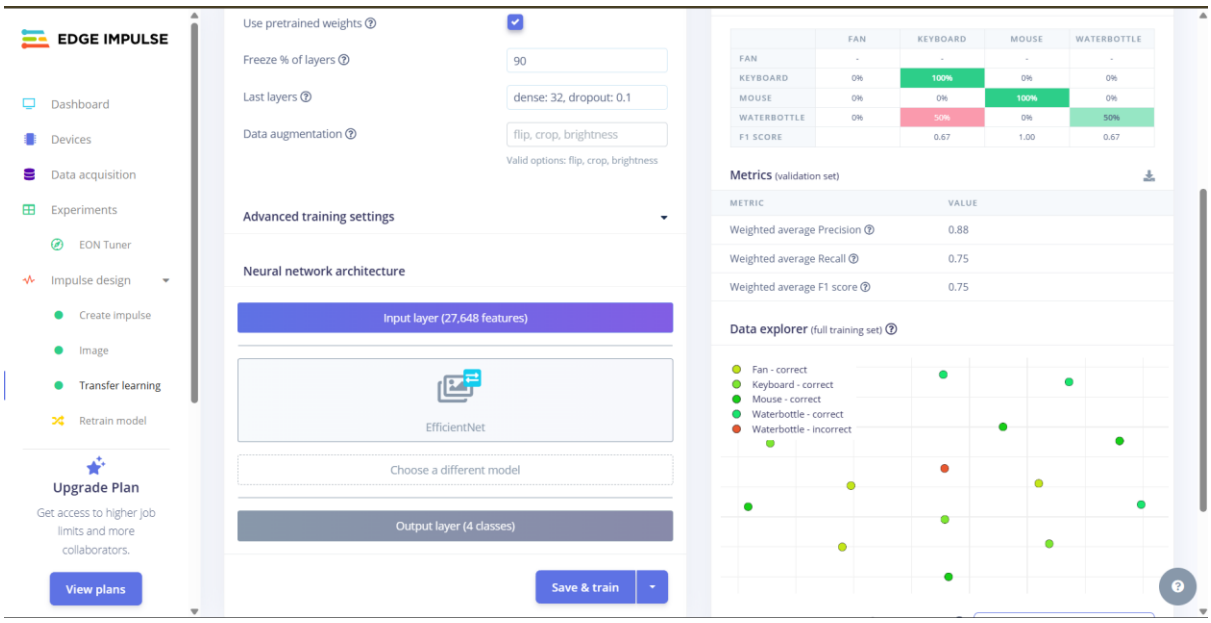
Now create the impulse by adding the processing block and learning block and save impulse

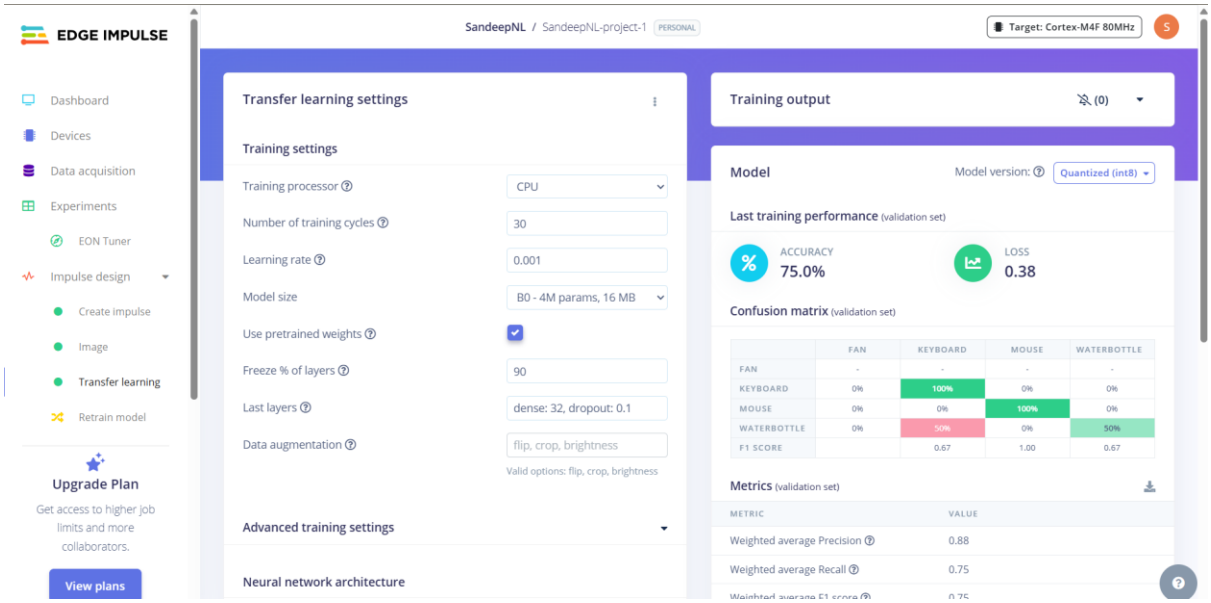


Now go to image tab and generate features



In the transfer learning choose one model from the given options and click train & save option





At last test the model to find the accuracy and implement with some other models to find the best fit model which gives the better accuracy output.